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Oefeningenreeks 1E nr. 26

$$A(z) = z^4 - 3z^3 + az^2 + bz - 8$$

$$A(z) = (z-2)^2(z^2 + pz + q)$$

$$\Rightarrow A(z) = (z^2 - 4z + 4)(z^2 + pz + q)$$

$$\Rightarrow A(z) = z^4 + (p-4)z^3 + (q - 4p + 4)z^2 + (-4q + 4p)z + 4q$$

$$\Rightarrow \begin{cases} p - 4 = -3 \\ q - 4p + 4 = a \\ -4q + 4p = b \\ 4q = -8 \end{cases}$$
$$\Rightarrow \begin{cases} p - 4 = -3 \\ q = -2 \end{cases}$$
$$\Rightarrow \begin{cases} p = 1 \\ q = -2 \end{cases}$$
$$\Rightarrow a = q - 4p + 4 = -2 - 4 + 4 = -2$$

$$\boxed{a = -2}$$

$$\Rightarrow b = -4q + 4p = 8 + 4 = 12$$

$$\boxed{b = 12}$$

References